

HEET MEHTA

New York, NY

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Portfolio

EDUCATION

New York University, Tandon School of Engineering

Sep 2025 to May 2027

Master of Science in Computer Science, **GPA: 3.83/4.0**

Brooklyn, NY

Courses: Software Engineering, Design & Analysis of Algorithms, Machine Learning, Big Data, Operating Systems

Vellore Institute of Technology

Sep 2021 to Jun 2025

Bachelor of Technology in Information Technology, **GPA: 9.14/10**

Tamil Nadu, India

Courses: Data Structures and Algorithms, Operating Systems, Computer Networks, Machine Learning, Database Management, Artificial Intelligence, Cloud Computing, Backend Development

SKILLS

Languages : Go, Python, Java, C++, TypeScript, JavaScript, SQL, Bash

Distributed Systems : Concurrency, Leader Election, High Availability, Event-Driven Architecture, WebSockets, gRPC

Backend and Infrastructure : REST APIs, PostgreSQL, SQLite, Kafka, Redis, Docker, Kubernetes, Bazel, Helm

Frontend : React, Next.js, Tailwind CSS, D3.js, Vite, Real-Time Data Visualization

AI and ML : PyTorch, Scikit-learn, LLM Integration, LangChain, Prompt Engineering, Kalman Filtering

Cloud and DevOps : AWS, GCP, CI/CD, GitHub Actions, Testing, Code Reviews, Technical Documentation

EXPERIENCE

Google (Intrinsic) | Software Engineering Intern, Data Infrastructure

May 2026 to Aug 2026

- Designed and delivered an end-to-end benchmarking toolchain in Go to evaluate metrics backend candidates for a production robot fleet platform, comparing Prometheus, VictoriaMetrics, Thanos, and Mimir under cardinality loads exceeding 782k time series.
- Engineered a containerized load generator built as a static distroless image via Bazel and deployed to Kubernetes with Helm, enabling reproducible validation of ingestion rate, memory consumption, and scrape latency across configurations.
- Authored a structured benchmark methodology and technical deliverable report synthesizing results for senior engineers, directly informing a data-driven infrastructure migration decision.

Bhabha Atomic Research Centre | Project Intern

Dec 2024 to May 2025

- Built a real-time sensor data pipeline in Python processing streams from 500+ IoT devices using multithreaded sockets; implemented RSSI-based trilateration with Kalman filtering to achieve 1-meter indoor positioning accuracy.
- Developed automated simulation scripts to validate pipeline behavior at scale, reducing evaluation time by 30%.

Encardio Rite | Software Development Intern

May 2024 to Aug 2024

- Built an ML-powered infrastructure diagnostics tool using Python and Scikit-learn deployed on AWS Lambda, reducing critical system failures by 40% across 20+ monitoring installations.
- Integrated sensor data from field devices via MQTT and AWS IoT Core; designed preprocessing pipelines and real-time dashboards that surfaced actionable insights to engineering teams.

PROJECTS

Aegis: Distributed Job Queue and Workflow Orchestration Engine | Go, PostgreSQL, React, TypeScript, WebSockets [Link](#)

- Designed and built a self-hosted, open-source alternative to Celery, Sidekiq, and Temporal: a distributed job queue and DAG workflow orchestrator in Go, implementing atomic job leasing, exponential backoff with jitter, dead-letter queues, and Postgres advisory lock based leader election for multi-node high availability.
- Built a storage abstraction layer supporting SQLite and PostgreSQL backends with race-free concurrent job claiming using transactional row locking (SELECT FOR UPDATE SKIP LOCKED), and a custom WebSocket protocol for real-time job dispatch with Go and Python client SDKs.
- Built a reactive DAG workflow engine supporting fan-out/fan-in dependency graphs and cycle detection driven by an internal event bus, paired with a real-time React and TypeScript dashboard featuring a custom SVG-based DAG visualizer.
- Packaged the entire system as a single deployable binary using Go's embed package, with Docker Compose configurations for single-node and multi-node HA deployments and GitHub Actions CI/CD.

SentryFlow: Real-Time API Rate Limiting and Analytics | Python, FastAPI, Redis, Kafka, ClickHouse [Link](#) 2025

- Deployed a production-grade platform on AWS handling 1000+ API requests daily with sub-100ms latency; implemented token bucket and sliding window algorithms with JWT authentication and a Kafka-backed event ingestion pipeline, maintaining 92% test coverage.
- Built an interactive React dashboard for real-time usage visualization following structured code review practices.

Eco-Nexus: AI Multi-Agent Procurement Platform | Next.js, TypeScript, Node.js, Snowflake, Socket.io [Link](#) 2025

- Developed an agentic AI negotiation engine evaluating 30+ supplier proposals with explainable scoring across cost, delivery, and carbon metrics, improving decision accuracy by 25%; real-time state sync via Socket.io supported 50+ concurrent sessions. Awarded 3rd Place at HackNYU 2026.

PUBLICATIONS

- Published “*Applications of Machine Learning in Detecting Unethical Sources of Raw Materials in Supply Chains in the Cosmetic Industry*” at **ICSES 2024, IEEE Xplore**.  [Link](#)

LEADERSHIP & SOCIAL IMPACT

- **Volunteer Web Developer, Catchafire**  [Link](#) — Led pro bono website redesign supporting workforce reintegration programs, improving navigation and accessibility for community impact.
- **Technical Team Lead, VIT Robotics Club** — Mentored 8 members across software and ML projects, organized 5 technical workshops on engineering best practices.